

Unit 5

Ethical & Legal Issues in the Digital Economy

Framework Library · 20 reference sheets

5.1**Data Rights & Privacy**

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LEGEND

DEFINITION	Conceptual definitions and key terms (8 cards)
FRAMEWORK	Strategic frameworks and analytical tools (12 cards)
K	

GDPR Principles

Seven principles · eight rights — the EU's foundational data-protection regime.

PROFESSIONAL DEFINITION

EU Regulation 2016/679 — the General Data Protection Regulation — establishes a rights-based regime for the processing of personal data in the EU. It rests on seven processing principles (lawfulness, purpose limitation, data minimisation, accuracy, storage limitation, integrity and confidentiality, accountability) and eight data-subject rights, with extraterritorial reach where EU residents' data is processed.

WHO DEVELOPED IT

Regulation (EU) 2016/679, applicable since 25 May 2018.

WHY IT WAS DEVELOPED

Successor to Directive 95/46/EC. Conceptual roots: Council of Europe Convention 108 (1981), OECD Privacy Guidelines (1980), EU Charter of Fundamental Rights Article 8.

FIGURE 1 · GDPR PRINCIPLES



Figure 1. GDPR Principles: Seven principles · eight rights — the EU's foundational data-protection regime.

HOW IT IS USED

when designing any product or system processing personal data.

WHEN IT IS USED

in privacy-by-design reviews and DPIAs.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Data Rights & Privacy.

REAL-WORLD EXAMPLES

- Meta WhatsApp 2023: €1.2bn fine for unlawful US data transfers.
- Google 2019: €50m CNIL fine for invalid consent for ads personalisation.
- TikTok 2023: €345m fine for processing children's data unlawfully.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, GDPR Principles contributes a clearer, more transferable framing of the dynamic it names — Seven principles · eight rights — the EU's foundational data-protection regime. Its analytical contribution is the explicit recognition that seven principles (Art. 5): lawfulness · purpose limitation · data minimisation · accuracy · storage limitation · integrity & confidentiality · accountability.

RELATED FRAMEWORKS IN THIS COURSE

- Personal Data as Property · 5.1.2
- Right to be Forgotten · 5.1.5
- DSA / DMA (Unit 3.4)

SOURCES (HARVARD REFERENCING STYLE)

European Union (2016) Regulation (EU) 2016/679 — General Data Protection Regulation. Brussels: Official Journal of the EU.

Voigt, P. and von dem Bussche, A. (2017) The EU General Data Protection Regulation (GDPR): A Practical Guide. Cham: Springer.

European Data Protection Board (2024) Annual Report 2024. Brussels: EDPB.

Personal Data as Property

SLIBRARY 17 · SHEET
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Is personal data my property — or is it relational, and beyond markets?

PROFESSIONAL DEFINITION

The contested philosophical and policy question: should personal data be treated as a property right (alienable, tradeable, owned by the data subject) or as a fundamental right (inalienable, contextual, beyond contractual transfer)? The EU framework leans firmly toward the rights model; some US scholarship and policy proposals lean toward property.

WHO DEVELOPED IT

Property model: Lessig, L. (1999) 'Code and Other Laws of Cyberspace'; Posner & Weyl (2018) 'Radical Markets'.

WHY IT WAS DEVELOPED

Rights model: Solove, D. (2008) 'Understanding Privacy'; EU Charter Article 8. Empirical critique in Acquisti, Taylor & Wagman (2016).

FIGURE 1 · PERSONAL DATA AS PROPERTY

Property vs. rights model		
Ownership	Subject owns	No owner; relational
Transfer	Sellable, licensable	Consent revocable
Limits	Markets + contract	Some uses prohibited
EU vs US	US/contract-leaning	EU/rights-leaning

Figure 1. Personal Data as Property: Is personal data my property — or is it relational, and beyond markets?.

HOW IT IS USED

when designing data-monetisation strategies — clarifies legal-philosophical limits.

WHEN IT IS USED

in policy and regulatory advocacy.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Data Rights & Privacy.

REAL-WORLD EXAMPLES

- Personal-data marketplaces (Datacoup, Mozilla data-broker experiments) — limited traction.
- Solid (Berners-Lee, 2018+): user-controlled 'pods' sit between property and rights.
- EU Data Act 2025: hybrid — IoT-data sharing rights without full propertisation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Personal Data as Property contributes a clearer, more transferable framing of the dynamic it names — Is personal data my property — or is it relational, and beyond markets?. Its analytical contribution is the explicit recognition that property model: data is owned by the subject; can be sold, traded, licensed; markets reveal price.

RELATED FRAMEWORKS IN THIS COURSE

[GDPR Principles · 5.1.1](#)
[Surveillance Capitalism — Zuboff Definition · 5.3.1](#)
[Data as Strategic Asset \(Unit](#)

SOURCES (HARVARD REFERENCING STYLE)

Posner, E.A. and Weyl, E.G. (2018) *Radical Markets: Uprooting Capitalism and Democracy for a Just Society*. Princeton: Princeton University Press.

Solove, D.J. (2008) *Understanding Privacy*. Cambridge, MA: Harvard University Press.

Acquisti, A., Taylor, C. and Wagman, L. (2016) 'The economics of privacy', *Journal of Economic Literature*, 54(2), pp. 442–492.

Consent & Legitimate Interest

Two of the six lawful bases — and the most contested in practice.

PROFESSIONAL DEFINITION

GDPR Article 6 lists six lawful bases for processing personal data; the two most heavily relied upon for digital-marketing and platform contexts are consent (Art. 6(1)(a)) and legitimate interest (Art. 6(1)(f)). Consent must be freely given, specific, informed, and unambiguous; legitimate interest requires a balancing test against the data subject's rights.

WHO DEVELOPED IT

Regulation (EU) 2016/679, Articles 6–7.

WHY IT WAS DEVELOPED

EDPB Guidelines 05/2020 on consent; EDPB Guidelines 01/2024 on legitimate interest.

FIGURE 1 · CONSENT & LEGITIMATE INTEREST



Figure 1. Consent & Legitimate Interest: Two of the six lawful bases — and the most contested in practice.

HOW IT IS USED

when designing consent flows for any EU-facing digital product.

WHEN IT IS USED

in legal review of personalised advertising and analytics.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Data Rights & Privacy.

REAL-WORLD EXAMPLES

- Meta 2023 (CJEU): cannot rely on legitimate interest or contract for ad personalisation — must obtain consent.
- Cookie-banner enforcement (CNIL France 2022+): €100m+ fines for invalid consent flows.
- TikTok 2023: €345m fine partly for invalid legitimate-interest reliance for processing children's data.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Consent & Legitimate Interest contributes a clearer, more transferable framing of the dynamic it names — Two of the six lawful bases — and the most contested in practice. Its analytical contribution is the explicit recognition that consent: must be (i) freely given, (ii) specific, (iii) informed, (iv) unambiguous; pre-ticked boxes invalid; revocable as easily as given.

RELATED FRAMEWORKS IN THIS COURSE

- GDPR Principles · 5.1.1
- Hyper-Personalisation (Unit 4.4.3)
- EU AI Act · 5.4.1

SOURCES (HARVARD REFERENCING STYLE)

- European Data Protection Board (2020) Guidelines 05/2020 on consent under Regulation 2016/679. Brussels: EDPB.
- European Data Protection Board (2024) Guidelines 01/2024 on processing of personal data based on legitimate interests. Brussels: EDPB.
- Court of Justice of the European Union (2023) Case C-252/21 Meta Platforms Inc. v Bundeskartellamt.

Data Minimisation

SLIBRARY 17 · SHEET
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Collect only what you need · keep only as long as you need · the under-rated principle.

PROFESSIONAL DEFINITION

The GDPR principle that personal data must be 'adequate, relevant and limited to what is necessary in relation to the purposes for which they are processed' (Art. 5(1)(c)). Combined with storage limitation (Art. 5(1)(e)), it forces firms to collect less and delete sooner than commercial intuition suggests.

WHO DEVELOPED IT

Regulation (EU) 2016/679, Article 5(1)(c).

WHY IT WAS DEVELOPED

Conceptual roots in 'privacy by design' (Cavoukian, 2011) and the 'least-privilege principle' from information security.

FIGURE 1 · DATA MINIMISATION

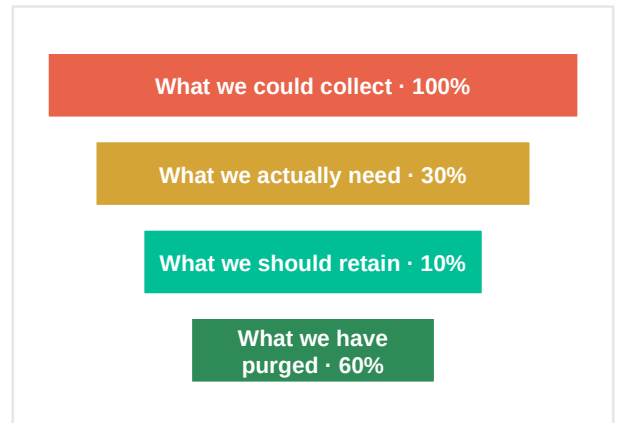


Figure 1. Data Minimisation: Collect only what you need · keep only as long as you need · the under-rated principle.

HOW IT IS USED

in DPIAs and privacy-by-design reviews.

WHEN IT IS USED

when scoping new data-collection features.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Data Rights & Privacy.

REAL-WORLD EXAMPLES

- Apple App Tracking Transparency (2021): operationalises minimisation by default for cross-app tracking.
- DuckDuckGo: search engine deliberately not retaining user search histories — minimisation as differentiation.
- Many financial firms 2023+: retention-period audits driven by GDPR enforcement focus.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Data Minimisation contributes a clearer, more transferable framing of the dynamic it names — Collect only what you need · keep only as long as you need · the under-rated principle. Its analytical contribution is the explicit recognition that three operational tests: (i) is each data field necessary for the purpose? (ii) is the granularity higher than needed? (iii) is the retention period longer than needed?

RELATED FRAMEWORKS IN THIS COURSE

GDPR Principles · 5.1.1

Right to be Forgotten · 5.1.5

Privacy Equation · 5.3.5

SOURCES (HARVARD REFERENCING STYLE)

Cavoukian, A. (2011) Privacy by Design: The 7 Foundational Principles. Toronto: Information and Privacy Commissioner of Ontario.

European Union (2016) Regulation (EU) 2016/679, Article 5. Brussels: Official Journal of the EU.

European Data Protection Board (2020) Guidelines 4/2019 on Article 25 — Data Protection by Design and by Default. Brussels: EDPB.

Right to be Forgotten

SLIBRARY 17 · SHEET
5 / 5

GDPR Art. 17 — the data subject's right to have personal data erased.

PROFESSIONAL DEFINITION

GDPR Article 17 creates a right for the data subject to obtain the erasure of personal data concerning them, where one of six specified grounds applies (e.g. data no longer necessary, consent withdrawn, unlawful processing). The right is qualified — not absolute — and subject to derogations including freedom of expression, public interest, and legal obligations.

WHO DEVELOPED IT

Conceptual roots in continental European 'droit à l'oubli'.

WHY IT WAS DEVELOPED

Operationalised by CJEU *Google Spain v AEPD* (Case C-131/12, 2014). Codified in GDPR Article 17 (2016/679).

FIGURE 1 · RIGHT TO BE FORGOTTEN

	Consent withdrawn	Unlawful processing
Erasure right	Right applies	Right applies
Derogation	—	—

Figure 1. Right to be Forgotten: GDPR Art. 17 — the data subject's right to have personal data erased.

HOW IT IS USED

when handling user erasure requests.

WHEN IT IS USED

in search-engine and content-platform legal operations.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Data Rights & Privacy.

REAL-WORLD EXAMPLES

- *Google Spain v AEPD* (CJEU 2014): de-listing right established.
- *Google v CNIL* (CJEU 2019): de-listing geographic scope limited to EU domains.
- *ND v Google* (Italy 2020+): right-to-be-forgotten extending into algorithmic outputs and historic press content.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Right to be Forgotten contributes a clearer, more transferable framing of the dynamic it names — GDPR Art. 17 — the data subject's right to have personal data erased. Its analytical contribution is the explicit recognition that six grounds for erasure (Art. 17(1)): no longer necessary · consent withdrawn · object successfully · unlawful processing · legal obligation · child-related collection.

RELATED FRAMEWORKS IN THIS COURSE

[GDPR Principles · 5.1.1](#)
[Data Minimisation · 5.1.4](#)
[DSA](#)

SOURCES (HARVARD REFERENCING STYLE)

European Court of Justice (2014) Case C-131/12 *Google Spain SL v Agencia Española de Protección de Datos*. Luxembourg: ECJ.

European Union (2016) Regulation (EU) 2016/679, Article 17. Brussels: Official Journal of the EU.

Mayer-Schönberger, V. (2009) *Delete: The Virtue of Forgetting in the Digital Age*. Princeton: Princeton University Press.

Copyright in the Digital Age

SLIBRARY 18 · SHEET
1 / 5*Pre-digital law · post-digital reality — the regime under maximum stress.*

PROFESSIONAL DEFINITION

Copyright is the exclusive legal right to reproduce, distribute, perform, display, or license a creative work for a limited period. In the digital age the regime faces structural strain: zero-cost copying, instant global distribution, AI training on copyrighted works, and the platform-as-distributor question all challenge a regime designed for printing presses and broadcasters.

WHO DEVELOPED IT

International framework: Berne Convention (1886, last revised 1971), TRIPS Agreement (1994), WIPO Copyright Treaty (1996).

WHY IT WAS DEVELOPED

EU: Information Society Directive 2001/29/EC, Copyright Directive 2019/790. US: Copyright Act 1976 + DMCA 1998.

FIGURE 1 · COPYRIGHT IN THE DIGITAL AGE

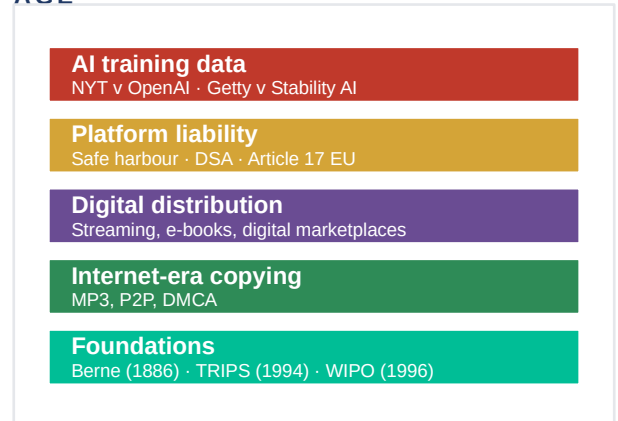


Figure 1. Copyright in the Digital Age: Pre-digital law · post-digital reality — the regime under maximum stress.

HOW IT IS USED

when assessing IP risk in any digital business model.

WHEN IT IS USED

in M&A diligence on creative or AI-training-data assets.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Intellectual Property.

REAL-WORLD EXAMPLES

- NYT v OpenAI (2023+): pending litigation on use of NYT articles in LLM training.
- Getty Images v Stability AI (2023+): pending US/UK cases on diffusion-model training.
- EU Copyright Directive Art. 17 (2019): platform-filtering obligations for UGC.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Copyright in the Digital Age contributes a clearer, more transferable framing of the dynamic it names — Pre-digital law · post-digital reality — the regime under maximum stress. Its analytical contribution is the explicit recognition that exclusive rights: reproduction · distribution · public performance · public display · derivative works · communication to the public.

RELATED FRAMEWORKS IN THIS COURSE

Creative Commons · 5.2.2

AI & Copyright · 5.2.4

Patent Trolls · 5.2.3

SOURCES (HARVARD REFERENCING STYLE)

World Intellectual Property Organization (1996) WIPO Copyright Treaty. Geneva: WIPO.

European Union (2019) Directive (EU) 2019/790 on Copyright and Related Rights in the Digital Single Market. Brussels: Official Journal of the EU.

Goldstein, P. and Hugenholtz, B. (2019) International Copyright: Principles, Law, and Practice. 4th edn. Oxford: Oxford University Press.

Creative Commons

A spectrum of standardised, machine-readable licences for permissive sharing.

PROFESSIONAL DEFINITION

Creative Commons (CC) is a non-profit organisation that publishes a suite of standardised licences allowing creators to grant the public specific permissions to use copyrighted works, while retaining other rights. Six main licence types span from 'CC BY' (attribution only) to 'CC BY-NC-ND' (most restrictive — no commercial use, no derivatives).

WHO DEVELOPED IT

Founded 2001 by Lawrence Lessig, Hal Abelson and Eric Eldred.

WHY IT WAS DEVELOPED

Lessig's intellectual context: 'Code and Other Laws of Cyberspace' (1999) and 'Free Culture' (2004). First licences released 2002.

FIGURE 1 · CREATIVE COMMONS

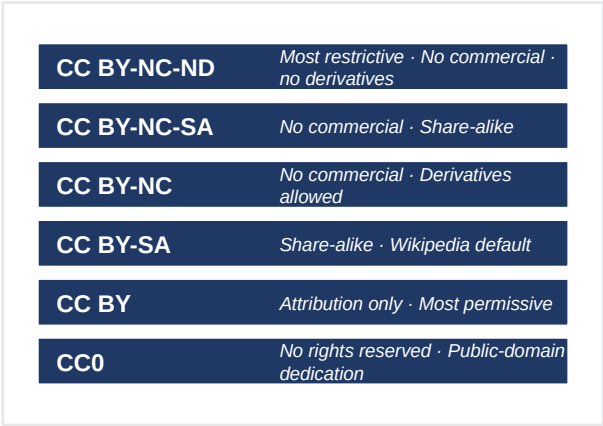


Figure 1. Creative Commons: A spectrum of standardised, machine-readable licences for permissive sharing.

HOW IT IS USED

when choosing how to license your own creative output.

WHEN IT IS USED

when sourcing content or training data with licensing constraints.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Intellectual Property.

REAL-WORLD EXAMPLES

- Wikipedia: CC BY-SA — share-alike requirement creates the 'copyleft' obligation.
- OpenStreetMap: ODbL (similar share-alike database licence).
- Government open data (data.gov, data.europa.eu): typically CC BY.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Creative Commons contributes a clearer, more transferable framing of the dynamic it names — A spectrum of standardised, machine-readable licences for permissive sharing. Its analytical contribution is the explicit recognition that six licences combining four optional elements: BY (attribution) · NC (non-commercial) · ND (no derivatives) · SA (share-alike).

RELATED FRAMEWORKS IN THIS COURSE

- Copyright in the Digital Age · 5.2.1
- AI & Copyright · 5.2.4
- Trade Secrets vs Patents · 5.2.5

SOURCES (HARVARD REFERENCING STYLE)

Lessig, L. (2004) Free Culture. New York: Penguin Press.

Creative Commons (2024) State of the Commons 2024. Mountain View, CA: Creative Commons.

Elkin-Koren, N. (2005) 'What contracts cannot do: the limits of private ordering in facilitating a Creative Commons', Fordham Law Review, 74, pp. 375–422.

Patent Trolls

Non-practising entities that hold patents to litigate, not to produce.

PROFESSIONAL DEFINITION

Non-practising entities (NPEs), colloquially 'patent trolls', acquire patents not to commercialise the underlying invention but to assert them against producing firms — extracting licensing fees through litigation threats. Particularly prevalent in software and digital-product industries, where patent quality is uneven and litigation costs are high.

WHO DEVELOPED IT

Term coined ~2001 (Peter Detkin, Intel).

WHY IT WAS DEVELOPED

Academic frame in Bessen, J. and Meurer, M. (2008) 'Patent Failure'. Reform debate intensified post-eBay v MercExchange (US 2006) and Alice v CLS Bank (US 2014).

FIGURE 1 · PATENT TROLLS



Figure 1. Patent Trolls: Non-practising entities that hold patents to litigate, not to produce.

HOW IT IS USED

in IP risk assessment for digital products.

WHEN IT IS USED

in policy advocacy on patent reform.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Intellectual Property.

REAL-WORLD EXAMPLES

- Intellectual Ventures: largest NPE; thousands of patents; multi-billion-dollar revenue.
- VirnetX v Apple (settled 2024): NPE-style litigation extracting hundreds of millions.
- Alice Corp v CLS Bank (US 2014): narrowed software-patent eligibility.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Patent Trolls contributes a clearer, more transferable framing of the dynamic it names — Non-practising entities that hold patents to litigate, not to produce. Its analytical contribution is the explicit recognition that nPE acquires patents — often broad, often weakly examined — from inventors or distressed firms.

RELATED FRAMEWORKS IN THIS COURSE

- Trade Secrets vs Patents · 5.2.5
- Copyright in the Digital Age · 5.2.1
- AI & Copyright · 5.2.4

SOURCES (HARVARD REFERENCING STYLE)

Bessen, J. and Meurer, M.J. (2008) Patent Failure: How Judges, Bureaucrats, and Lawyers Put Innovators at Risk. Princeton: Princeton University Press.

Lemley, M.A. and Melamed, A.D. (2013) 'Missing the forest for the trolls', Columbia Law Review, 113, pp. 2117–2189.

Cohen, L., Gurun, U.G. and Kominers, S.D. (2019) 'Patent trolls: evidence from targeted firms', Management Science, 65(12), pp. 5461–5486.

AI & Copyright

Three open questions · no settled answers · billion-dollar litigation pending.

PROFESSIONAL DEFINITION

The frontier IP question of the 2020s: how does copyright law apply to (i) the use of copyrighted works in AI training; (ii) the copyrightability of AI-generated outputs; (iii) the liability for AI outputs that reproduce protected works. Active litigation in the US, UK, EU, and China; no settled jurisprudence as of 2025.

WHO DEVELOPED IT

Major test cases: New York Times v OpenAI (US 2023+), Getty Images v Stability AI (US/UK 2023+), Authors Guild v OpenAI (US 2023+).

WHY IT WAS DEVELOPED

EU regulatory frame: AI Act (2024) plus Copyright Directive Art. 4 (text and data mining exception). Academic foundation: Sag, M. (2023) 'Copyright safety for generative AI', Houston Law Review.

FIGURE 1 · AI & COPYRIGHT

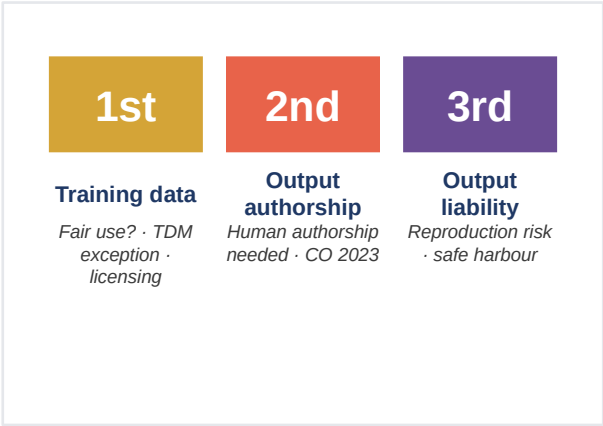


Figure 1. AI & Copyright: Three open questions · no settled answers · billion-dollar litigation pending.

HOW IT IS USED

in legal review of any generative-AI deployment.

WHEN IT IS USED

in licensing strategy — both as licensor (data owner) and licensee (AI firm).

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Intellectual Property.

REAL-WORLD EXAMPLES

- NYT v OpenAI (filed Dec 2023): allegations of verbatim article reproduction.
- Getty Images v Stability AI: similar allegations re: image diffusion training.
- US Copyright Office (2023): refused registration of AI-generated images.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, AI & Copyright contributes a clearer, more transferable framing of the dynamic it names — Three open questions · no settled answers · billion-dollar litigation pending. Its analytical contribution is the explicit recognition that question 1 — Training data: is using copyrighted works to train AI a fair use / fair dealing? US: contested; EU: TDM exception with opt-out; Japan: explicit safe harbour.

RELATED FRAMEWORKS IN THIS COURSE

- Copyright in the Digital Age · 5.2.1
- Creative Commons · 5.2.2
- EU AI Act · 5.4.1

SOURCES (HARVARD REFERENCING STYLE)

Sag, M. (2023) 'Copyright safety for generative AI', Houston Law Review, 61, pp. 295–375.

European Union (2019) Directive (EU) 2019/790, Article 4 — text and data mining exception. Brussels: Official Journal of the EU.

US Copyright Office (2023) Copyright Registration Guidance: Works Containing Material Generated by Artificial Intelligence. Washington, DC: USCO.

Trade Secrets vs Patents

SLIBRARY 18 · SHEET
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Two strategies for protecting innovation · radically different trade-offs.

PROFESSIONAL DEFINITION

Trade secrets and patents are alternative legal regimes for protecting commercially valuable information. Patents grant time-limited monopolies in exchange for public disclosure; trade secrets protect indefinitely but only as long as secrecy is maintained. The choice depends on reverse-engineerability, technology lifespan, and litigation appetite.

WHO DEVELOPED IT

Patents: international framework via Paris Convention (1883) and TRIPS (1994).

WHY IT WAS DEVELOPED

Trade secrets: US Defend Trade Secrets Act (2016); EU Trade Secrets Directive (2016). Strategic comparison foundational in Friedman, Landes & Posner (1991).

FIGURE 1 · TRADE SECRETS VS PATENTS

Patents vs. trade secrets		
Duration	20 years	Indefinite if secret
Disclosure	Public disclosure	Strict secrecy
Protection	Independent reinvention covered	Independent reinvention not covered
Reverse engineering	Risk after disclosure	Lawful · undermines protection

Figure 1. Trade Secrets vs Patents: Two strategies for protecting innovation · radically different trade-offs.

HOW IT IS USED

when designing IP strategy for a technology asset.

WHEN IT IS USED

in M&A diligence — IP composition affects valuation and integration.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Intellectual Property.

REAL-WORLD EXAMPLES

● Coca-Cola: trade secret since 1886 — outlived multiple patent terms.

● Pharmaceutical formulas: typically patented (FDA disclosure required).

● Algorithm trade secrets: Google ranking, Facebook News Feed — too valuable to patent.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Trade Secrets vs Patents contributes a clearer, more transferable framing of the dynamic it names — Two strategies for protecting innovation · radically different trade-offs. Its analytical contribution is the explicit recognition that patents: 20-year monopoly · public disclosure required · enforceable against independent invention · jurisdictional.

RELATED FRAMEWORKS IN THIS COURSE

[Patent Trolls · 5.2.3](#)
[Copyright in the Digital Age · 5.2.1](#)
[Data as Strategic Asset \(Unit](#)

SOURCES (HARVARD REFERENCING STYLE)

Friedman, D.D., Landes, W.M. and Posner, R.A. (1991) 'Some economics of trade secret law', *Journal of Economic Perspectives*, 5(1), pp. 61–72.

European Union (2016) Directive (EU) 2016/943 on the Protection of Undisclosed Know-How and Business Information (Trade Secrets). Brussels: Official Journal of the EU.

Lemley, M.A. (2008) 'The surprising virtues of treating trade secrets as IP rights', *Stanford Law Review*, 61, pp. 311–353.

Zuboff's Definition

SLIBRARY 19 · SHEET
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A new economic logic · human experience as raw material for behavioural prediction products.

PROFESSIONAL DEFINITION

Shoshana Zuboff defines surveillance capitalism as a new economic logic that 'unilaterally claims human experience as free raw material for translation into behavioural data', subsequently fabricating 'prediction products' that are sold in 'behavioural futures markets'. Distinguishable from earlier surveillance regimes by its commercial logic and instrumentarian power.

WHO DEVELOPED IT

Zuboff, S. (2019) 'The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power'.

WHY IT WAS DEVELOPED

Earlier articulations: Zuboff, S. (2015) 'Big other: surveillance capitalism and the prospects of an information civilization', Journal of Information Technology.

FIGURE 1 · ZUBOFF'S DEFINITION



Figure 1. Zuboff's Definition: A new economic logic · human experience as raw material for behavioural prediction products.

HOW IT IS USED

when assessing the structural ethics of any data-monetising business model.

WHEN IT IS USED

in policy discussions on platform regulation.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Surveillance Capitalism.

REAL-WORLD EXAMPLES

- Google: archetypal surveillance-capitalism firm — Zuboff's 'mother lode'.
- Meta: Cambridge Analytica (2018) revealed political-modification dimension.
- Pokémon Go (Niantic): real-world location-data extraction with engagement design.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Zuboff's Definition contributes a clearer, more transferable framing of the dynamic it names — A new economic logic · human experience as raw material for behavioural prediction products. Its analytical contribution is the explicit recognition that step 1 — Extraction: human experience (clicks, location, biometrics, voice, browsing) extracted as 'behavioural surplus'.

RELATED FRAMEWORKS IN THIS COURSE

Behavioural Futures Markets · 5.3.2

Data Extraction Imperative · 5.3.3

Instrumentarian Power · 5.3.4

SOURCES (HARVARD REFERENCING STYLE)

Zuboff, S. (2019) The Age of Surveillance Capitalism. New York: PublicAffairs.

Zuboff, S. (2015) 'Big other: surveillance capitalism and the prospects of an information civilization', Journal of Information Technology, 30(1), pp. 75–89.

Foucault, M. (1977) Discipline and Punish: The Birth of the Prison. New York: Pantheon.

Behavioural Futures Markets

SLIBRARY 19 · SHEET
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Markets in predictions of what humans will do · the product is human behaviour.

PROFESSIONAL DEFINITION

Markets in which sellers offer probabilistic predictions of human behaviour — clicks, purchases, votes, churn, default risk — and buyers pay for the predictions to inform decisions. The 'commodity' is statistical claims about people; participation is typically non-consensual and the data subject is rarely the beneficiary.

WHO DEVELOPED IT

Zuboff (2019) names the concept; analytical foundations in attention-economy literature (Wu, 2016) and information-asymmetry economics (Akerlof, 1970).

WHY IT WAS DEVELOPED

The framework was developed to give practitioners a precise, transferable vocabulary for the dynamic captured in its tagline — replacing the looser pre-existing terms used in earlier industry and academic discourse.

FIGURE 1 · BEHAVIOURAL FUTURES MARKETS



Figure 1. Behavioural Futures Markets: Markets in predictions of what humans will do · the product is human behaviour.

HOW IT IS USED

when classifying a data-monetising business model.

WHEN IT IS USED

in regulatory analysis of advertising and credit-scoring markets.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Surveillance Capitalism.

REAL-WORLD EXAMPLES

- Google Ad Exchange: real-time behavioural-futures market for click-prediction.
- Meta Audience Network: similar logic, applied across the Meta ecosystem and partners.
- Credit-scoring firms (Experian, Equifax): predictions of repayment behaviour sold to lenders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Behavioural Futures Markets contributes a clearer, more transferable framing of the dynamic it names — Markets in predictions of what humans will do · the product is human behaviour. Its analytical contribution is the explicit recognition that data assemblers extract behavioural traces from users.

RELATED FRAMEWORKS IN THIS COURSE

[Zuboff's Definition · 5.3.1](#)
[Data Extraction Imperative · 5.3.3](#)
[Hyper-Personalisation \(Unit 4.](#)

SOURCES (HARVARD REFERENCING STYLE)

Zuboff, S. (2019) *The Age of Surveillance Capitalism*. New York: PublicAffairs.

Wu, T. (2016) *The Attention Merchants*. New York: Knopf.

Akerlof, G.A. (1970) 'The market for lemons: quality uncertainty and the market mechanism', *Quarterly Journal of Economics*, 84(3), pp. 488–500.

Data Extraction Imperative

SLIBRARY 19 · SHEET
3 / 5

Once the data flywheel turns · stopping is uncompetitive · extraction expands.

PROFESSIONAL DEFINITION

The structural pressure on surveillance-capitalist firms to continuously expand the volume, granularity, and reach of data collection. Stopping (or even slowing) extraction is competitively disadvantageous because rivals will out-predict, out-target, and out-monetise. The result is a one-way ratchet of expanding extraction.

WHO DEVELOPED IT

Zuboff (2019), drawing on industrial-era 'imperative' analyses (e.g.

WHY IT WAS DEVELOPED

Marx's analysis of capital's tendency to expand). Empirically grounded by privacy-economics studies (Acquisti, Taylor & Wagman, 2016).

FIGURE 1 · DATA EXTRACTION IMPERATIVE

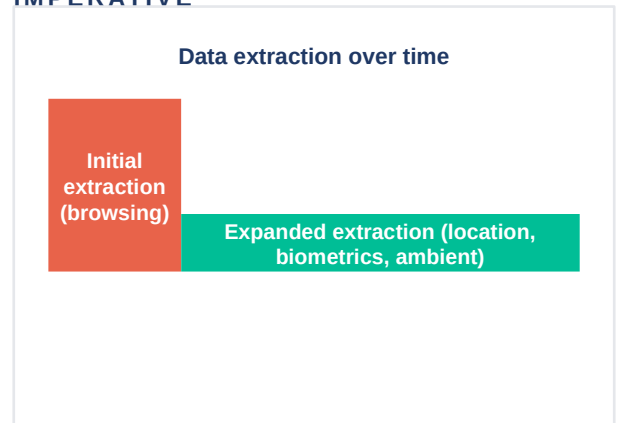


Figure 1. Data Extraction Imperative: Once the data flywheel turns · stopping is uncompetitive · extraction expands.

HOW IT IS USED

in structural critique of platform business models.

WHEN IT IS USED

in regulatory advocacy and data-minimisation enforcement.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Surveillance Capitalism.

REAL-WORLD EXAMPLES

- Smartphones (2007–): extraction expanded from web browsing to location, contacts, app usage.
- Smart speakers / voice assistants: extraction extended into the domestic acoustic environment.
- Connected cars (2020–): extraction extended to driving behaviour, in-cabin video.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Data Extraction Imperative contributes a clearer, more transferable framing of the dynamic it names — Once the data flywheel turns · stopping is uncompetitive · extraction expands. Its analytical contribution is the explicit recognition that volume imperative: more data → more accurate predictions → more value.

RELATED FRAMEWORKS IN THIS COURSE

[Zuboff's Definition · 5.3.1](#)
[Behavioural Futures Markets · 5.3.2](#)
[Data Minimisation · 5.1.4](#)

SOURCES (HARVARD REFERENCING STYLE)

- Zuboff, S. (2019) *The Age of Surveillance Capitalism*. New York: PublicAffairs.
- Acquisti, A., Taylor, C. and Wagman, L. (2016) 'The economics of privacy', *Journal of Economic Literature*, 54(2), pp. 442–492.
- Couldry, N. and Mejias, U.A. (2019) *The Costs of Connection*. Stanford: Stanford University Press.

Instrumentarian Power

SLIBRARY 19 · SHEET
4 / 5

Zuboff — power that shapes behaviour from the outside · without coercion · without consent.

PROFESSIONAL DEFINITION

Shoshana Zuboff's term for the distinctive form of power exercised by surveillance-capitalist firms: not totalitarian coercion (forcing behaviour under threat) and not classical disciplinary power (changing souls), but instrumentarian power — shaping behaviour from outside through nudges, defaults, recommendations, and feedback systems, while leaving the inner self apparently untouched.

WHO DEVELOPED IT

Zuboff (2019) 'Age of Surveillance Capitalism', drawing on but distinguishing from Foucault's 'biopower' and Hannah Arendt's analysis of totalitarianism.

WHY IT WAS DEVELOPED

Earlier behavioural-economics framing in Thaler & Sunstein (2008) 'Nudge'.

FIGURE 1 · INSTRUMENTARIAN POWER

Forms of power		
Totalitarian	Coerces from outside	Threat of force
Disciplinary	Reshapes the soul	Internalised norm
Instrumentarian	Shapes behaviour from outside	Nudge, ranking, default
Visibility	Hidden from subject	Visible to firm only

Figure 1. Instrumentarian Power: Zuboff — power that shapes behaviour from the outside · without coercion · without consent.

HOW IT IS USED

in ethical analysis of UX design choices (dark patterns, nudges, defaults).

WHEN IT IS USED

in regulatory and academic analysis of platform power.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Surveillance Capitalism.

REAL-WORLD EXAMPLES

- Infinite scroll, autoplay, push notifications: classic instrumentarian shaping.
- Algorithmic recommendations: shape what is seen, considered, chosen.
- Dark patterns in consent flows: pre-ticked boxes, confirm-shaming, hidden opt-outs.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Instrumentarian Power contributes a clearer, more transferable framing of the dynamic it names — Zuboff — power that shapes behaviour from the outside · without coercion · without consent. Its analytical contribution is the explicit recognition that operates through behavioural modification: defaults, frictions, ranking, recommendation, gamification, feedback.

RELATED FRAMEWORKS IN THIS COURSE

Zuboff's Definition · 5.3.1

Hyper-Personalisation (Unit 4.

Algorithmic Bias · 5.4.4

SOURCES (HARVARD REFERENCING STYLE)

Zuboff, S. (2019) The Age of Surveillance Capitalism. New York: PublicAffairs.

Thaler, R.H. and Sunstein, C.R. (2008) Nudge: Improving Decisions about Health, Wealth, and Happiness. New Haven: Yale University Press.

Foucault, M. (1977) Discipline and Punish: The Birth of the Prison. New York: Pantheon.

Privacy Equation (MIT)

MIT IDE — privacy is a continuously-renegotiated social contract, not a fixed state.

PROFESSIONAL DEFINITION

MIT IDE's framing of privacy as an ongoing negotiation rather than a fixed property: $\text{privacy}(t) = \text{function of (data flowing in, social norms, regulatory layer, user agency at time } t\text{)}$. The implication: privacy outcomes depend more on continuous design choices and regulatory pressure than on one-time consent or static policy.

WHO DEVELOPED IT

MIT Initiative on the Digital Economy (2025) IDE 2025 Annual Report.

WHY IT WAS DEVELOPED

Conceptual lineage: Helen Nissenbaum (2010) 'Privacy in Context' (contextual integrity); Solove (2008) 'Understanding Privacy' (taxonomy of privacy harms).

FIGURE 1 · PRIVACY EQUATION (MIT)

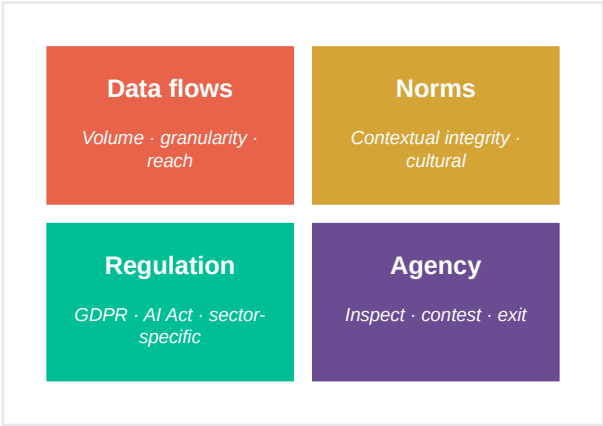


Figure 1. Privacy Equation (MIT): MIT IDE — privacy is a continuously-renegotiated social contract, not a fixed state.

HOW IT IS USED

when designing privacy reviews that go beyond compliance.

WHEN IT IS USED

in academic and policy analysis of privacy outcomes.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Surveillance Capitalism.

REAL-WORLD EXAMPLES

- Apple App Tracking Transparency (2021): boosted user agency leg.
- GDPR (2018): boosted regulation leg significantly.
- Smart speakers: data-flow leg expanded faster than norms or regulation could keep up.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Privacy Equation (MIT) contributes a clearer, more transferable framing of the dynamic it names — MIT IDE — privacy is a continuously-renegotiated social contract, not a fixed state. Its analytical contribution is the explicit recognition that $\text{privacy} = f(\text{data flows, norms, regulation, agency})$ — continuously updated, never fixed.

RELATED FRAMEWORKS IN THIS COURSE

- GDPR Principles · 5.1.1
- Data Minimisation · 5.1.4
- Zuboff's Definition · 5.3.1

SOURCES (HARVARD REFERENCING STYLE)

MIT Initiative on the Digital Economy (2025) IDE 2025 Annual Report. Cambridge, MA: MIT IDE.

Nissenbaum, H. (2010) Privacy in Context: Technology, Policy, and the Integrity of Social Life. Stanford: Stanford University Press.

Solove, D.J. (2008) Understanding Privacy. Cambridge, MA: Harvard University Press.

EU AI Act

EU Regulation 2024/1689 — the world's first comprehensive risk-based AI regulation.

PROFESSIONAL DEFINITION

EU Regulation 2024/1689 establishes a horizontal, risk-based regulatory regime for AI systems placed on the EU market or affecting EU persons. AI systems are classified into four risk tiers: unacceptable (banned), high-risk (heavily regulated), limited-risk (transparency duties), and minimal-risk (unregulated). General-purpose AI models are covered separately.

WHO DEVELOPED IT

Regulation (EU) 2024/1689, adopted June 2024; phased applicability through 2025–2027.

WHY IT WAS DEVELOPED

Drafted by DG-CNECT 2021–2024. Conceptual roots: EC High-Level Expert Group on AI (2019) Ethics Guidelines.

FIGURE 1 · EU AI ACT

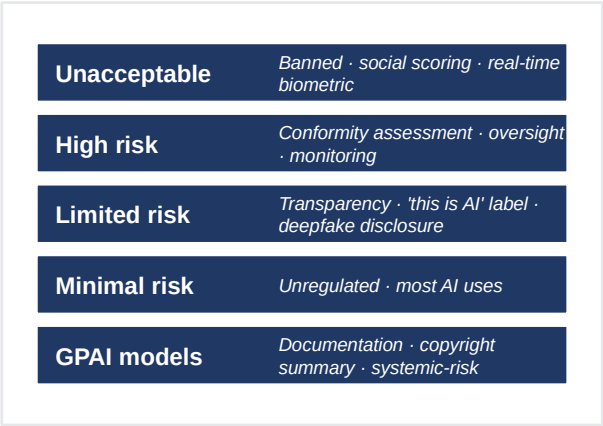


Figure 1. EU AI Act: EU Regulation 2024/1689 — the world's first comprehensive risk-based AI regulation.

HOW IT IS USED

in AI-governance design for any EU-facing AI system.

WHEN IT IS USED

in product launches involving AI features.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Algorithmic Accountability.

REAL-WORLD EXAMPLES

- Banned: real-time biometric ID in public spaces (narrow law-enforcement exceptions).
- High-risk: AI for hiring, credit, education examination — full conformity regime.
- GP AI: GPT-4, Claude, Gemini — model documentation, copyright summary, risk evaluation.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, EU AI Act contributes a clearer, more transferable framing of the dynamic it names — EU Regulation 2024/1689 — the world's first comprehensive risk-based AI regulation. Its analytical contribution is the explicit recognition that tier 1 — Unacceptable risk (banned): social scoring, real-time biometric ID in public, manipulative or exploitative AI.

RELATED FRAMEWORKS IN THIS COURSE

- Algorithmic Bias · 5.4.4
- Responsibility Gap · 5.4.3
- GDPR Principles · 5.1.1

SOURCES (HARVARD REFERENCING STYLE)

European Union (2024) Regulation (EU) 2024/1689 — Artificial Intelligence Act. Brussels: Official Journal of the EU.

Veale, M. and Zuiderveen Borgesius, F. (2021) 'Demystifying the Draft EU AI Act', Computer Law Review International, 22(4), pp. 97–112.

European AI Office (2024) Implementing the AI Act: Guidelines and Codes of Practice. Brussels: European Commission.

Accountability Spectrum

SLIBRARY 20 · SHEET
2 / 5

From mere conduit to active gatekeeper — five positions of platform responsibility.

PROFESSIONAL DEFINITION

A spectrum of platform responsibility positions, ranging from passive 'mere conduit' (no editorial control, full safe harbour) through 'host' and 'curator' to 'publisher' (full editorial responsibility). Where a platform sits on this spectrum determines its legal liability for user-generated and AI-generated content.

WHO DEVELOPED IT

Conceptual lineage: e-Commerce Directive 2000/31/EC (Articles 12–15) and US CDA Section 230.

WHY IT WAS DEVELOPED

Modernised in Klonick (2018) 'New Governors', Harvard Law Review, and codified by EU DSA (2022) tier system.

FIGURE 1 · ACCOUNTABILITY SPECTRUM

5 · Publisher	Edits / creates · Full liability
4 · Active curator	Ranks / recommends · Contested safe harbour
3 · Hosting	Stores user content · Safe harbour on notice
2 · Caching	Temporary storage · Conditional safe harbour
1 · Mere conduit	Transmission only · Full safe harbour

Figure 1. Accountability Spectrum: From mere conduit to active gatekeeper — five positions of platform responsibility.

HOW IT IS USED

when classifying a platform's legal exposure for content.

WHEN IT IS USED

in T&S strategy and intermediary-liability litigation.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Algorithmic Accountability.

REAL-WORLD EXAMPLES

● Telecoms: mere conduit — no liability for transmitted content.

● Pre-2017 YouTube: hosting — relied on notice-and-takedown safe harbour.

● Modern YouTube / TikTok: arguably active curator — algorithmic ranking expands liability.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Accountability Spectrum contributes a clearer, more transferable framing of the dynamic it names — From mere conduit to active gatekeeper — five positions of platform responsibility. Its analytical contribution is the explicit recognition that position 1 — Mere conduit: no knowledge, no control over transmission. Full safe harbour (e-Commerce Directive Art. 12).

RELATED FRAMEWORKS IN THIS COURSE

DSA

Platform Governance (Unit 3.3.)

Responsibility Gap · 5.4.3

SOURCES (HARVARD REFERENCING STYLE)

European Union (2000) Directive 2000/31/EC on Electronic Commerce. Brussels: Official Journal of the EU.

Klonick, K. (2018) 'The new governors: the people, rules, and processes governing online speech', Harvard Law Review, 131(6), pp. 1598–1670.

Husovec, M. (2024) Principles of the Digital Services Act. Oxford: Oxford University Press.

Responsibility Gap

SLIBRARY 20 · SHEET
3 / 5

Andreas Matthias — when no human can fairly be held accountable for a machine's harm.

PROFESSIONAL DEFINITION

The responsibility gap is the gap between the harm caused by an autonomous AI system and the persons who can plausibly be held morally and legally responsible. As AI systems become more autonomous and adaptive, traditional fault attribution to designers, deployers, or users becomes strained — creating a normative and legal void.

WHO DEVELOPED IT

Matthias, A. (2004) 'The responsibility gap: ascribing responsibility for the actions of learning automata', Ethics and Information Technology.

WHY IT WAS DEVELOPED

Refined by Sparrow (2007), Nyholm (2018), and integrated into AI-policy discourse by HLEG-AI (2019).

FIGURE 1 · RESPONSIBILITY GAP

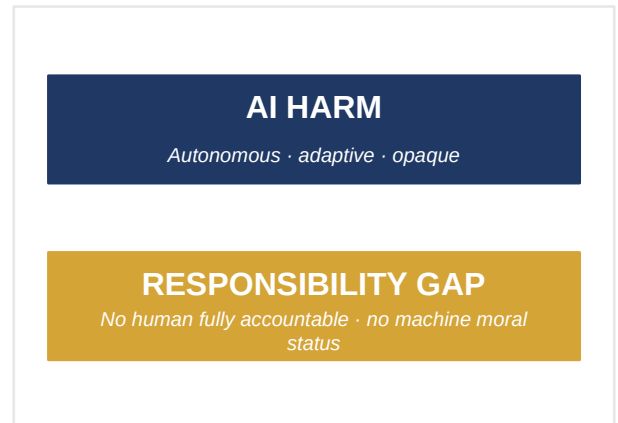


Figure 1. Responsibility Gap: Andreas Matthias — when no human can fairly be held accountable for a machine's harm.

HOW IT IS USED

in ethics and governance reviews for autonomous AI systems.

WHEN IT IS USED

in liability and insurance design.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Algorithmic Accountability.

REAL-WORLD EXAMPLES

● Tesla Autopilot crashes (2016+): allocation of responsibility extensively litigated.

● Wrongful arrest (Detroit 2020): facial recognition false positive.

● AI Act response: high-risk systems require explicit human oversight.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Responsibility Gap contributes a clearer, more transferable framing of the dynamic it names — Andreas Matthias — when no human can fairly be held accountable for a machine's harm. Its analytical contribution is the explicit recognition that designers: cannot foresee all behaviours of a learning system.

RELATED FRAMEWORKS IN THIS COURSE

[EU AI Act · 5.4.1](#)[Algorithmic Bias · 5.4.4](#)[AI Risk & Mitigation Matrix · 5.4.5](#)

SOURCES (HARVARD REFERENCING STYLE)

Matthias, A. (2004) 'The responsibility gap: ascribing responsibility for the actions of learning automata', Ethics and Information Technology, 6(3), pp. 175–183.

Sparrow, R. (2007) 'Killer robots', Journal of Applied Philosophy, 24(1), pp. 62–77.

Nyholm, S. (2018) 'Attributing agency to automated systems: reflections on human-robot collaborations and responsibility-loci', Science and Engineering Ethics, 24(4), pp. 1201–1219.

Algorithmic Bias

SLIBRARY 20 · SHEET
4 / 5

Three sources · three responses · the central operational risk in AI deployment.

PROFESSIONAL DEFINITION

Algorithmic bias is the systematic error in algorithmic decisions that disadvantages specific groups, typically along protected characteristics (race, gender, age, disability). Sources include biased training data, proxy variables, mis-specified objectives, and feedback loops. Bias has been documented in hiring, lending, criminal justice, and healthcare AI.

WHO DEVELOPED IT

Foundational empirical work: Buolamwini, J. and Gebru, T. (2018) 'Gender Shades' (PMLR).

WHY IT WAS DEVELOPED

Comprehensive academic treatment: Barocas, S., Hardt, M. and Narayanan, A. (2023) 'Fairness and Machine Learning'. Earlier ProPublica analysis of COMPAS recidivism scoring (Angwin et al., 2016).

FIGURE 1 · ALGORITHMIC BIAS

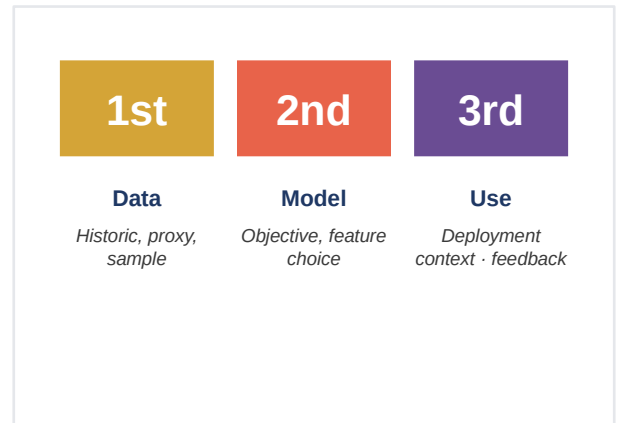


Figure 1. Algorithmic Bias: Three sources · three responses · the central operational risk in AI deployment.

HOW IT IS USED

in AI-system design and audit.

WHEN IT IS USED

in regulatory compliance (EU AI Act, US Fair Lending).

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Algorithmic Accountability.

REAL-WORLD EXAMPLES

- ProPublica COMPAS analysis (2016): recidivism algorithm flagged Black defendants disproportionately.
- Gender Shades (2018): face-recognition error gap up to 35% across demographic subgroups.
- Amazon hiring tool (2018, withdrawn): systematically penalised female-coded CV terms.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Algorithmic Bias contributes a clearer, more transferable framing of the dynamic it names — Three sources · three responses · the central operational risk in AI deployment. Its analytical contribution is the explicit recognition that source 1 — Data: historical bias in training data (hiring records reflecting historic discrimination); proxy variables that correlate with protected attributes.

RELATED FRAMEWORKS IN THIS COURSE

EU AI Act · 5.4.1

Responsibility Gap · 5.4.3

AI Risk & Mitigation Matrix · 5.4.5

SOURCES (HARVARD REFERENCING STYLE)

Barocas, S., Hardt, M. and Narayanan, A. (2023) Fairness and Machine Learning. Cambridge, MA: MIT Press.

Buolamwini, J. and Gebru, T. (2018) 'Gender shades: intersectional accuracy disparities in commercial gender classification', PMLR, 81, pp. 77–91.

Mehrabi, N., Morstatter, F., Saxena, N., Lerman, K. and Galstyan, A. (2021) 'A survey on bias and fairness in machine learning', ACM Computing Surveys, 54(6), pp. 1–35.

AI Risk & Mitigation Matrix (MIT)

SLIBRARY 20 · SHEET 5 / 5

MIT IDE 2025 — a 2x2 of AI risk types and mitigation maturity.

PROFESSIONAL DEFINITION

MIT IDE's 2025 framework for classifying AI deployment risk along two axes: type of risk (operational vs. societal) and mitigation maturity (immature vs. mature). The four quadrants imply different governance approaches — engineering controls, model audits, regulatory compliance, multi-stakeholder oversight.

WHO DEVELOPED IT

MIT Initiative on the Digital Economy (2025) IDE 2025 Annual Report.

WHY IT WAS DEVELOPED

Builds on NIST AI Risk Management Framework (2023) and EU AI Act risk tiers.

FIGURE 1 · AI RISK & MITIGATION MATRIX (MIT)

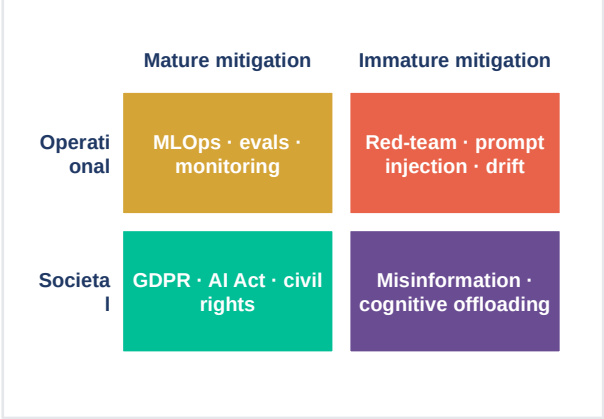


Figure 1. AI Risk & Mitigation Matrix (MIT): MIT IDE 2025 — a 2x2 of AI risk types and mitigation maturity.

HOW IT IS USED

in AI-portfolio risk assessment.

WHEN IT IS USED

in board-level AI-governance reviews.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Algorithmic Accountability.

REAL-WORLD EXAMPLES

- Operational/mature: production recommender systems — well-instrumented MLOps.
- Operational/immature: agentic AI tool-use — red-teaming, eval frameworks emerging.
- Societal/immature: large-scale persuasion / microtargeting — limited governance toolkit.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, AI Risk & Mitigation Matrix (MIT) contributes a clearer, more transferable framing of the dynamic it names — MIT IDE 2025 — a 2x2 of AI risk types and mitigation maturity. Its analytical contribution is the explicit recognition that operational + mature mitigation: model accuracy, latency, robustness — addressed via MLOps, eval suites.

RELATED FRAMEWORKS IN THIS COURSE

- EU AI Act · 5.4.1
- Algorithmic Bias · 5.4.4
- Cognitive Trade-Off (Unit 4.3.

SOURCES (HARVARD REFERENCING STYLE)

MIT Initiative on the Digital Economy (2025) IDE 2025 Annual Report. Cambridge, MA: MIT IDE.

NIST (2023) AI Risk Management Framework (AI RMF 1.0). Gaithersburg, MD: National Institute of Standards and Technology.

European Union (2024) Regulation (EU) 2024/1689 — Artificial Intelligence Act. Brussels: Official Journal of the EU.